

FLUTTER ☐ ☐ ☐ ☐

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2-4 /



Flutter

1. Widget Element RenderObject
2. setState
3. const
4. GlobalKey



- 5.
6. 2MB OOM
7. Future microtask isolate
- 8.



9. Riverpod/Bloc/GetX
10. App
- 11.
- 12.



13. WebSocket
- 14.
15. WebSocket



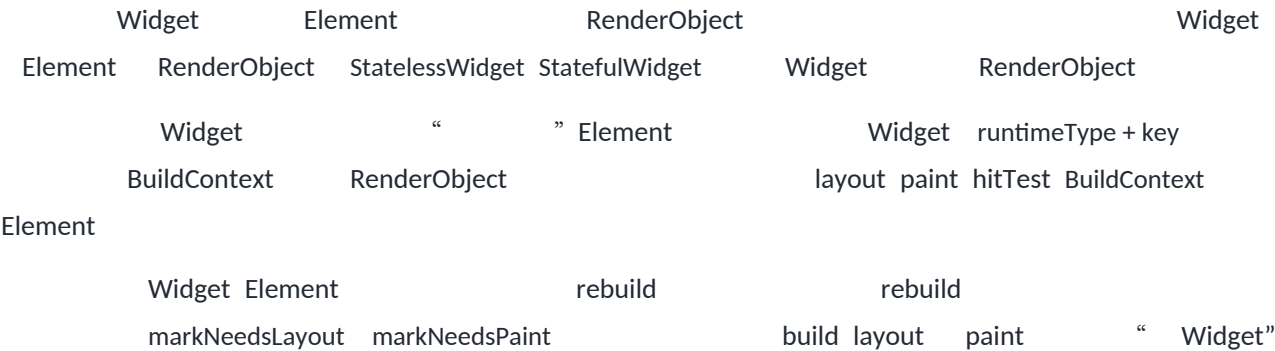
- 16.
17. MethodChannel EventChannel
18. Android/iOS
- 19.



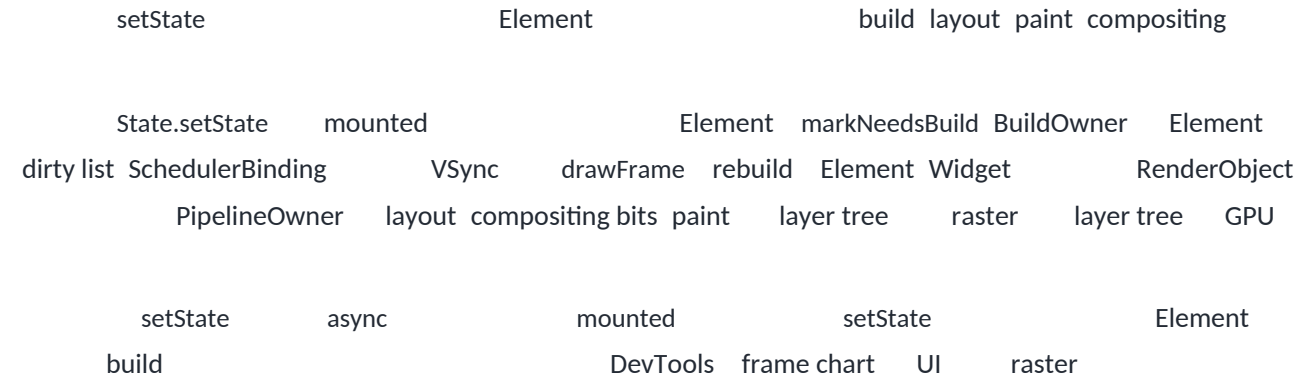
- 20.



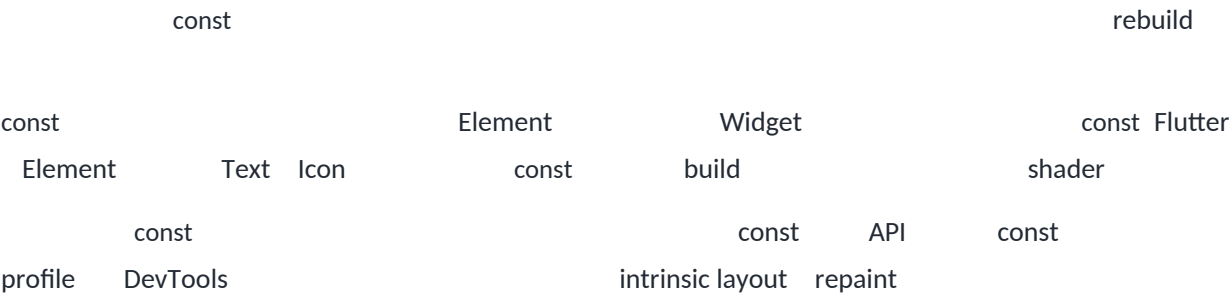
1. Widget Element RenderObject



2. setState



3. const



4. GlobalKey



FormState.validate()	ScaffoldMessenger	State	
reparent	deactivate/activate	InheritedWidget	GlobalKey
build	GlobalKey		key
GlobalKey	State	“ ”	ValueKey(id)
LayoutBuilder		GlobalKey	Controller



5.

60Hz 16.7ms 120Hz 8.3ms UI build/layout raster
shader
profile DevTools Performance frame chart UI jank raster jank rebuild stats Track
Widget Builds Repaint Rainbow CPU profiler memory/image cache UI builder
item build JSON/ / IO shrinkWrap IntrinsicHeight Raster
saveLayer repaint
ListView.builder/Sliver item key build repaint
RepaintBoundary P90/P99 frame time “ ”

6. 2MB OOM

2MB × × 4 bytes CPU GPU
8000×6000 JPEG RGBA 183MB GPU EXIF
ImageCache Dart heap Dart heap
DPR cacheWidth/cacheHeight ResizeImage “ 100px
4000px” DevTools Memory Android dumsys
meminfo/Profiler iOS Instruments Dart heap native heap GPU OOM ImageCache

7. Future microtask isolate

Future microtask isolate event queue isolate
Future
async/await isolate microtask queue event timer IO microtask
microtask UI IO Future()/Timer event queue IO
await UI isolate
CPU JSON UI isolate Isolate.run worker isolate isolate
TransferableTypedData IO Future microtask CPU
isolate

8.

GC native

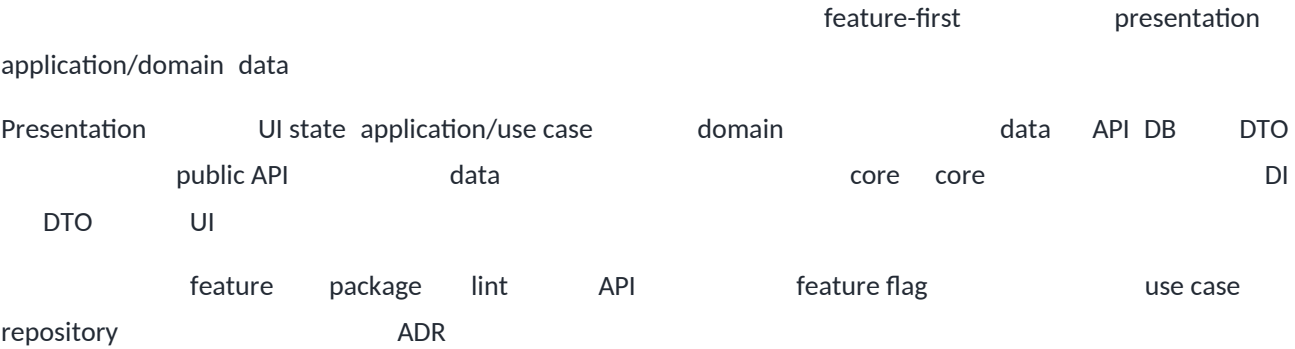
		20	profile	Dart heap	external/native memory	RSS	DevTools
Memory	diff snapshot	retained size		retaining path		StreamSubscription/Timer	
AnimationController/ScrollController		dispose		State	Provider	EventChannel	
observer/delegate							
	Dart heap	RSS	Android Studio Profiler	LeakCanary	dumpsys meminfo	iOS Instruments	
Leaks/Allocations		bitmap		GC	retained objects		
retaining path							



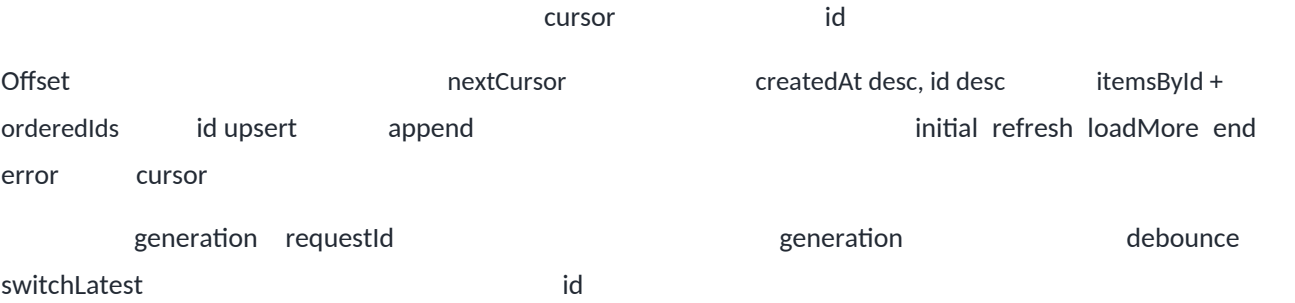
9. Riverpod/Bloc/GetX



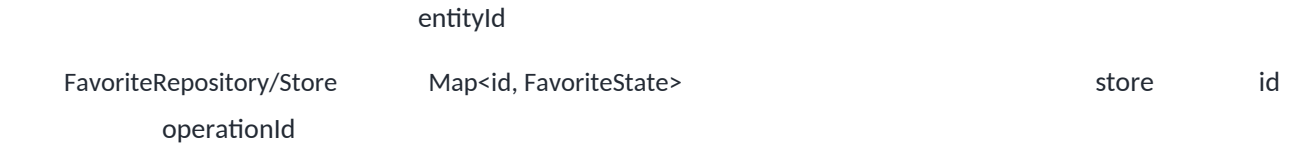
10. App



11.



12.



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PUT/DELETE toggle store /
WebSocket/ UI



13. WebSocket

WebSocket		“ ”		id		
channel/conversationId		messageId	serverSeq	serverTime	payload	messageId
lastAckSeq	resume token	lastSeq	seq	expectedSeq	/	
API	expectedSeq	ack				
clientMessageId		pending	sent	acknowledged	failed	id
WebSocket	REST	serverSeq				

14.

				TCP	
ping	pong	25	ping 10	pong	App
Android/iOS					
full jitter	30-60	401	token	connectivity	
generation	socket			ready	

15. WebSocket

WebSocket		App				
WebSocket	push	eventId	/	APNs/FCM	payload	event
		API	App	sync token		
messageld/eventld		socket	push			
token	key	iOS				



16.

	draftId	URI	hash	uploadId	multipart	hash/clientFileId	TTL
failed	2-3						
remoteKey							
	draftId/idempotencyKey		“ ”				
	token						

17. MethodChannel EventChannel

MethodChannel	-	EventChannel	Dart	codec
MethodChannel	RPC	Dart	Dart	SDK
EventChannel	Dart	Stream	onListen/onCancel	EventSink
	schema		Dart	EventChannel
listener			codec	FFI
				Federated plugin
				cancel

18. Android/iOS

	"	"				CI			
	feature flag		release	lint	/UI			Android	
versionCode	iOS build number		changelog	CI		Android AAB	iOS Archive/IPA		mapping.txt
dSYM									
Android	Play Console	/	Data safety		release notes	staged rollout	iOS	App Store	
Connect	TestFlight	/			phased release				crash-free
users	ANR	/	tag						

19.

		feature flag				
		1%	5%	20%	/	crash-free
ANR	OOM	P95/P99				trace breadcrumb
release id						

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flag
kill switch

expand-migrate-contract
“ ”



20.

STAR

Situation	Android	3-5	OOM	Crash	native allocation failed	Dart
	crash-free users	99.7%	98.9%			

Task

Action	DevTools	Dart heap	RSS	Graphics	GPU	dumpsys meminfo	Android
Profiler		120dp	4000px		ImageProvider		MB
	MB	CDN	URL	xDPR			
A/B	PSS	GPU	P99 frame time				
Result	20	PSS	620MB	260MB	P99	46ms	19ms
						crash-free	99.75%
						Dart/native/GPU	

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15	60-120	30-60	20
	“ ”	“ ”	